

ROS1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ROS1 gene fusions from the msk_impact_50k_2026 study, covering 122 fusion events. 72/122 fusions (59.0%) were in-frame. 107/122 fusions (87.7%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.214047$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for ROS1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 122 analyzed fusion events, identifying 30 distinct fusion partner genes. The most frequent partner was CD74, observed in 55 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 67/72 (93.1%) of in-frame fusions retained the domain, $p=0.214047$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Disruption of the Fibronectin type III domain was tested with Fisher's exact test comparing in-frame fusions against all others; 68/72 (94.4%) of in-frame fusions disrupted the domain, $p=0.0770779$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

Cutpoint detection

Cutpoint detection scanned 118 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1694 aa (permutation-corrected $p=0.178218$, not statistically significant at $\alpha=0.05$). This is 26 aa from the nearest configured domain boundary at 1668 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=107$) and "not_retained" ($n=11$). For Frame_status, retained: 67/90 (74.4%, 95% CI 64.6%-82.3%); not_retained: 5/9 (55.6%, 95% CI 26.7%-81.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 68.5514 vs 19.0909, $p=3.23984e-07$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (30 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: ROS1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CD74	55	0.45081967213114754	0.20043213737826426	0.0	0.9192443359110025	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.38713063229312644	1
SDC4	17	0.13934426229508196	0.20043213737826426	0.0	0.9547152194211018	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.29900864186882165	2
EZR	13	0.10655737704918032	0.20043213737826426	0.0	0.9195548041701888	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2838985140074142	3
COL13A1	1	0.00819672131147541	0.20043213737826426	0.0	1.0	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26645709666057443	4
ZNF157	1	0.00819672131147541	0.20043213737826426	0.0	1.0	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26645709666057443	5
AFG1L	1	0.00819672131147541	0.20043213737826426	0.0	0.9993894993894994	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26636552156899934	6
SLC34A2	1	0.00819672131147541	0.20043213737826426	0.0	0.9993894993894994	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26636552156899934	7
FN1	1	0.00819672131147541	0.20043213737826426	0.0	0.9658119658119658	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26132889153236927	8
RBPJL	1	0.00819672131147541	0.20043213737826426	0.0	0.9658119658119658	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26132889153236927	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NKAIN2	1	0.00819672131147541	0.20043213737826426	0.0	0.9633699633699634	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2609625911660689	10
STX7	1	0.00819672131147541	0.20043213737826426	0.0	0.942002442002442	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2577574629609407	11
GOPC	4	0.03278688524590164	0.20043213737826426	0.0	0.878968253968254	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25567938393614037	12
SLC4A4	2	0.01639344262295082	0.20043213737826426	0.0	0.9029304029304029	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25435567349357746	13
TFG	3	0.02459016393442623	0.20043213737826426	0.0	0.8858363858363858	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25425058732291755	14
IGF1R	1	0.00819672131147541	0.20043213737826426	0.0	0.9029304029304029	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25189665710013487	15
LARS1	1	0.00819672131147541	0.20043213737826426	0.0	0.9029304029304029	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25189665710013487	16
LRIG3	2	0.01639344262295082	0.20043213737826426	0.0	0.8858363858363858	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2517915709294749	17
NUDCD3	2	0.01639344262295082	0.20043213737826426	0.0	0.8858363858363858	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2517915709294749	18
TPM3	2	0.01639344262295082	0.20043213737826426	0.0	0.8858363858363858	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2517915709294749	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CCDC30	1	0.00819672131147541	0.20043213737826426	0.0	0.8858363858363858	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2493325545360323	20
GOLGB1	1	0.00819672131147541	0.20043213737826426	0.0	0.8858363858363858	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2493325545360323	21
MYH9	1	0.00819672131147541	0.20043213737826426	0.0	0.8583638583638584	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2452116754151532	22
PRKN	1	0.00819672131147541	0.20043213737826426	0.0	0.8388278388278388	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24228127248475023	23
CEP85L	2	0.01639344262295082	0.20043213737826426	0.0	0.7481684981684982	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23114138777929177	24
GRIK2	1	0.00819672131147541	0.20043213737826426	0.0	0.673992673992674	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21755599775947554	25
NUS1	1	0.00819672131147541	0.20043213737826426	0.0	0.22283272283272282	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.14988200508548286	26
CERT1	1	0.00819672131147541	0.20043213737826426	0.0	None	0.6389004592256574	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.1370083490124405	27
EYS	1	0.00819672131147541	0.20043213737826426	0.0	None	0.6389004592256574	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.1370083490124405	28
NETO1	1	0.00819672131147541	0.20043213737826426	0.0	None	0.6389004592256574	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.1370083490124405	29
SLC16A10	1	0.00819672131147541	0.20043213737826426	0.0	0.0	0.6389004592256574	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11645709666057442	30

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
56	0	107	1	10	0.0	0.09322033898305085	1.0304893221479003
421	1	106	1	10	0.09433962264150944	0.17847312762567	0.7484271655598992
875	2	105	1	10	0.19047619047619047	0.25637653793702886	0.5911217213946672
1430	3	104	1	10	0.28846153846153844	0.32750573865609567	0.4847810857642878
1634	4	103	1	10	0.3883495145631068	0.39239553580331443	0.4062759421356547
1688	5	102	1	10	0.49019607843137253	0.4515428730260006	0.34530100806444014
1693	7	100	1	10	0.7	0.554422700897893	0.2561589957973137
1694	8	99	3	8	0.21548821548821548	0.06612721119379524	1.1796197925642002
1750	26	81	4	7	0.5617283950617284	0.4675112851720896	0.3302079013073733
1755	27	80	4	7	0.590625	0.47622392260409285	0.3221887920600884
1789	44	63	6	5	0.582010582010582	0.5242960133693759	0.28042344455265833
1800	45	62	6	5	0.6048387096774194	0.527867316527827	0.27747522697617977
1840	47	60	6	5	0.6527777777777778	0.5392135184056008	0.2682392283136832
1853	91	16	10	1	0.56875	1.0	-0.0
1881	104	3	11	0	0.0	1.0	-0.0
1926	106	1	11	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

68	39
4	7

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00041	Fibronectin type III domain	118	11	0	107	0.0

domain_retention tables

frame_domain_contingency_table

67	40
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5	6
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domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	118	107	0	11	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
36	107	122	0.8770491803278688

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AFG1L	1
CCDC30	1
CD74	55
CEP85L	2
CERT1	1
COL13A1	1
EYS	1
EZR	13
FN1	1
GOLGB1	1
GOPC	4
GRIK2	1
IGF1R	1
LARS1	1
LRIG3	2
MYH9	1
NETO1	1
NKAIN2	1

Partner_gene	Event_count
NUDCD3	2
NUS1	1
PRKN	1
RBPJL	1
SDC4	17
SLC16A10	1
SLC34A2	1
SLC4A4	2
STX7	1
TFG	3
TPM3	2
ZNF157	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 7022-T01-I M6-1	P-0037 022-T0 1-IM6		AFG1 L-RO S1	AFG 1L	in-frame	three_prime	117660 863	30	1693	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase	Fibronectin type III domain		LACE1 (NM_145315) - ROS1 (NM_002944) rearrangement: c.808-19179 :LACE1_c.5079+1435:ROS1 inv Protein Fusion: in frame {LACE1:ROS1}	NA
EVT-P-004 1695-T01-I M6-3	P-0041 695-T0 1-IM6		CCD C30- ROS1	CC DC3 0	out-of-frame	three_prime	117643 359	35	1881	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase	Fibronectin type III domain		CCDC30 (NM_001080850) - ROS1 (NM_002944) fusion: t (1;6)(p34.2;q22.1)(chr1:g.43 087864::chr6:g.117643359) Protein Fusion: out of frame {CCDC30:ROS1}	NA
EVT-P-001 4506-T02-I M6-6	P-0014 506-T0 2-IM6		CD74 -ROS 1	CD7 4	unknown	three_prime	117650 623	32	1750	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t (5;6)(q32;q22.1)(chr5:g.1497 82872::chr6:g.117650623) Protein Fusion: mid-exon {CD74:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 1025-T02-I M6-7	P-0011 025-T0 2-IM6		CD74 -ROS 1	CD7 4	in-frame	threonine_prime	117645 889	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783756::chr6:g.117645889) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-005 4497-T01-I M6-8	P-0054 497-T0 1-IM6		CD74 -ROS 1	CD7 4	out-of-frame	threonine_prime	117659 373	31	1694	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783252::chr6:g.117659373) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-005 4440-T02-I M6-9	P-0054 440-T0 2-IM6		CD74 -ROS 1	CD7 4	in-frame	threonine_prime	117646 088	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782403::chr6:g.117646088) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-005 4352-T01-I M6-10	P-0054 352-T0 1-IM6		CD74 -ROS 1	CD7 4	out-of-frame	threonine_prime	117649 394	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783536::chr6:g.117649394) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-001 4506-T05-I M7-11	P-0014 506-T0 5-IM7		CD74 -ROS 1	CD7 4	unknown	threonine_prime	117650 623	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782872::chr6:g.117650623) Protein Fusion: mid-exon {CD74:ROS1}	NA
EVT-P-002 6455-T01-I M6-12	P-0026 455-T0 1-IM6		CD74 -ROS 1	CD7 4	in-frame	threonine_prime	117646 566	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783651::chr6:g.117646566) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-005 2363-T02-I M7-13	P-0052 363-T0 2-IM7		CD74 -ROS 1	CD7 4	in-frame	threonine_prime	117647 256	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783700::chr6:g.117647256) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 2363-T01-I M6-14	P-0052 363-T0 1-IM6		CD74-ROS1	CD74	in-frame	threonine_prime	117647256	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783692::chr6:g.117647256) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-004 9640-T03-I M7-15	P-0049 640-T0 3-IM7		CD74-ROS1	CD74	out-of-frame	threonine_prime	117650417	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782214::chr6:g.117650417) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-004 9640-T01-I M6-16	P-0049 640-T0 1-IM6		CD74-ROS1	CD74	out-of-frame	threonine_prime	117650417	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782214::chr6:g.117650417) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-004 8671-T02-I M6-17	P-0048 671-T0 2-IM6		CD74-ROS1	CD74	in-frame	threonine_prime	117647250	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783869::chr6:g.117647250) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-001 1025-T01-I M5-18	P-0011 025-T0 1-IM5		CD74-ROS1	CD74	in-frame	threonine_prime	117645889	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 34-43) : t(5;6)(q32;q22.1)(chr5:g.149783755::chr6:g.117645889) Protein fusion: in frame (CD74:ROS1)	NA
EVT-P-004 8671-T01-I M6-19	P-0048 671-T0 1-IM6		CD74-ROS1	CD74	in-frame	threonine_prime	117647250	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783869::chr6:g.117647250) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-004 7502-T01-I M6-20	P-0047 502-T0 1-IM6		CD74-ROS1	CD74	in-frame	threonine_prime	117646405	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782474::chr6:g.117646405) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 4465-T01-I M6-21	P-0044 465-T0 1-IM6		CD74-ROS1	CD74	in-frame	threonine_prime	117646909	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783638::chr6:g.117646909) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-000 0840-T03-I M6-22	P-0000 840-T0 3-IM6		CD74-ROS1	CD74	in-frame	threonine_prime	117646938	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783509::chr6:g.117646938) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-004 1118-T06-I M6-23	P-0041 118-T0 6-IM6		CD74-ROS1	CD74	unknown	threonine_prime	117647425	33	1840	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782265::chr6:g.117647425) Protein Fusion: mid-exon {CD74:ROS1}	NA
EVT-P-004 1118-T02-I M6-24	P-0041 118-T0 2-IM6		CD74-ROS1	CD74	unknown	threonine_prime	117647424	33	1840	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782265::chr6:g.117647424) Protein Fusion: mid-exon {CD74:ROS1}	NA
EVT-P-000 4747-T04-I M7-25	P-0004 747-T0 4-IM7		CD74-ROS1	CD74	out-of-frame	threonine_prime	117649107	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782562::chr6:g.117649107) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-004 0869-T01-I M6-26	P-0040 869-T0 1-IM6		CD74-ROS1	CD74	in-frame	threonine_prime	117647263	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783792::chr6:g.117647263) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-004 0577-T01-I M6-27	P-0040 577-T0 1-IM6		CD74-ROS1	CD74	out-of-frame	threonine_prime	117650159	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 32-43) : t(5;6)(q32;q22.1)(chr5:g.149783441::chr6:g.117650159) Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 9219-T01-I M6-28	P-0039 219-T0 1-IM6		CD74-ROS1	CD74	in-frame	threonine_prime	117646339	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783535:chr6:g.117646339) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-002 4572-T01-I M6-29	P-0024 572-T0 1-IM6		CD74-ROS1	CD74	out-of-frame	threonine_prime	117649758	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion(CD74 exons 1-6 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149783583:chr6:g.117649758) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-005 4667-T01-I M6-30	P-0054 667-T0 1-IM6		CEP85L-ROS1	CEP85L	unknown	threonine_prime	117662314	30	1688	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CEP85L (NM_001178035) - ROS1 (NM_002944) fusion: c.1149-2421:CEP85L_c.5063:ROS1del Protein Fusion: mid-exon {CEP85L:ROS1}	NA
EVT-P-006 3351-T02-I M7-33	P-0063 351-T0 2-IM7		EZR-ROS1	EZR	in-frame	threonine_prime	117645650	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1091-221:EZR_c.5558-72:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-000 3970-T02-I M5-34	P-0003 970-T0 2-IM5		EZR-ROS1	EZR	out-of-frame	threonine_prime	117649698	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: (EZR exons 1 to 9 fused ROS1 exons 33 to 43): c.1090+142:EZR_c.5366_794:ROS1del Protein fusion: out of frame (EZR-ROS1)	NA
EVT-P-002 0458-T04-I M6-35	P-0020 458-T0 4-IM6		EZR-ROS1	EZR	in-frame	threonine_prime	117646033	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1090+208:EZR_c.5558-455:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-002 8081-T01-I M6-36	P-0028 081-T0 1-IM6		EZR-ROS1	EZR	in-frame	threonine_prime	117647275	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1090+285:EZR_c.5557+112:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 8494-T01-I M7-38	P-0068 494-T0 1-IM7		FN1- ROS1	FN1	in-frame	threonine_prime	117652028	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		FN1 (NM_212482) - ROS1 (NM_002944) fusion: t(2;6)(q35;q22.1)(chr2:g.216267900::chr6:g.117652028) Protein Fusion: in frame {FN1:ROS1}	NA
EVT-P-002 1147-T01-I M6-40	P-0021 147-T0 1-IM6		GOPC- ROS1	GOPC	in-frame	threonine_prime	117641701	36	1926	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		GOPC (NM_020399) - ROS1 (NM_002944) fusion (GOPC exons 1-4 fused with ROS1 exons 36-43) :c.651-735:GOPC_c.5778-508:ROS1del Protein Fusion: in frame {GOPC:ROS1}	NA
EVT-P-001 5494-T01-I M6-44	P-0015 494-T0 1-IM6		IGF1R- ROS1	IGF1R	in-frame	threonine_prime	117646453	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		IGF1R (NM_000875) - ROS1 (NM_002944) Rearrangement : t(6;15)(q22.1,q26.3)(chr6:g.117646453::chr15:g.99477638) Protein fusion: in frame (IGF1R-ROS1)	NA
EVT-P-006 8483-T01-I M7-45	P-0068 483-T0 1-IM7		LRIG3- ROS1	LRIG3	in-frame	threonine_prime	117644742	34	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		LRIG3 (NM_153377) - ROS1 (NM_002944) fusion: t(6;12)(q22.1;q14.1)(chr6:g.117644742::chr12:g.59269495) Protein Fusion: in frame {LRIG3:ROS1}	NA
EVT-P-002 3007-T01-I M6-46	P-0023 007-T0 1-IM6		LRIG3- ROS1	LRIG3	in-frame	threonine_prime	117645256	34	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		LRIG3 (NM_153377) - ROS1 (NM_002944) fusion: (LRIG3 exons 1-15 fused to ROS1 exons 35 - 43): t(6;12)(q22.1;q14.1)(chr6:g.117645256::chr12:g.59270940) Protein Fusion: in frame {LRIG3:ROS1}	NA
EVT-P-002 8295-T01-I M6-47	P-0028 295-T0 1-IM6		MYH9- ROS1	MYH9	in-frame	threonine_prime	117641564	36	1926	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		MYH9 (NM_002473) - ROS1 (NM_002944) rearrangement: t(6;22)(q22.1;q12.3)(chr6:g.117641564::chr22:g.36691922) Protein Fusion: in frame {MYH9:ROS1}	NA
EVT-P-003 7864-T02-I M7-49	P-0037 864-T0 2-IM7		NKAIN2- ROS1	NKAIN2	unknown	threonine_prime	117662563	29	1634	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		NKAIN2 (NM_001040214) - ROS1 (NM_002944) fusion: c.55-122327:NKAIN2_c.4902:ROS1inv Protein Fusion: mid-exon {NKAIN2:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 4050-T03-I M6-50	P-0054 050-T0 3-IM6		NUDCD3-ROS1	NUDCD3	out-of-frame	threep_prime	117645481	34	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		NUDCD3 (NM_015332) - ROS1 (NM_002944) fusion: t(6;7)(q22.1;p13)(chr6:g.117645481::chr7:g.44523353) Protein Fusion: out of frame {NUDCD3:ROS1}	NA
EVT-P-005 4050-T05-I M7-51	P-0054 050-T0 5-IM7		NUDCD3-ROS1	NUDCD3	out-of-frame	threep_prime	117645481	34	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		NUDCD3 (NM_015332) - ROS1 (NM_002944) fusion: t(6;7)(q22.1;p13)(chr6:g.117645481::chr7:g.44523353) Protein Fusion: out of frame {NUDCD3:ROS1}	NA
EVT-P-006 7287-T01-I M7-52	P-0067 287-T0 1-IM7		PRKN-ROS1	PRKN	unknown	threep_prime	117674184	26	1430	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		PARK2 (NM_004562) - ROS1 (NM_002944) fusion: c.171+704:PARK2_c.4290:ROS1del Protein Fusion: mid-exon {PARK2:ROS1}	NA
EVT-P-001 9205-T02-I M6-53	P-0019 205-T0 2-IM6		RBPJL-ROS1	RBPJL	unknown	five_prime	117654098	32	1750	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		ROS1 (NM_002944) rearrangement: t(6;20)(q22.1;q13.12)(chr6:g.117654098::chr20:g.43949086) Antisense Fusion	NA
EVT-P-002 1228-T01-I M6-57	P-0021 228-T0 1-IM6		ROS1-CD74	CD74	in-frame	threep_prime	117646295	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783729::chr6:g.117646295) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-001 0760-T01-I M5-58	P-0010 760-T0 1-IM5		ROS1-CD74	CD74	in-frame	threep_prime	117647234	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(5q32;q22.1)(chr5:g.149783763::chr6:g.117647234) Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 2707-T03-I M5-59	P-0002 707-T0 3-IM5		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117646 946	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD274 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.49783748::chr6:g.117646946) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-001 5215-T01-I M6-60	P-0015 215-T0 1-IM6		ROS1 -CD7 4	CD7 4	out-of-frame	threonine_prime	117650 370	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 32-43): t(5;6)(q32;q22.1)(chr5:g.149783057::chr6:g.117650370) Protein fusion: out of frame (CD74-ROS1)	NA
EVT-P-000 2877-T01-I M3-61	P-0002 877-T0 1-IM3		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117645 870	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		NA Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-001 2525-T01-I M5-62	P-0012 525-T0 1-IM5		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117647 385	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1(NM_002944) fusion (CD74 exons 1-6 with ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783203::chr6:g.117647385) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-000 2912-T01-I M3-63	P-0002 912-T0 1-IM3		ROS1 -CD7 4	CD7 4	in-frame	five_prime	117646 326	34	1853	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		NA Protein fusion: in frame (ROS1-CD74)	NA
EVT-P-000 4336-T01-I M5-64	P-0004 336-T0 1-IM5		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117647 249	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 with ROS1 exons 34-43) : t(5;6)(5q32;q22.1)(chr5:g.149783071::chr6:g.117647249) Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 6213-T01-IM6-65	P-0016 213-T01-IM6		ROS1-CD74	CD74	out-of-frame	three_prime	117649786	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149783418::chr6:g.117649786) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-000 2707-T01-IM3-66	P-0002 707-T01-IM3		ROS1-CD74	CD74	in-frame	five_prime	117646946	33	1853	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		NA Protein fusion: in frame (ROS1-CD74)	NA
EVT-P-000 4747-T01-IM5-67	P-0004 747-T01-IM5		ROS1-CD74	CD74	out-of-frame	three_prime	117649107	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149782561::chr6:g.117649107) Protein fusion: out of frame (CD74-ROS1)	NA
EVT-P-000 4790-T01-IM5-68	P-0004 790-T01-IM5		ROS1-CD74	CD74	in-frame	three_prime	117646467	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (exons 1-7 of CD74 fused to exons 34-43 of ROS1) : t(5;6)(q32;q22.1)(chr5:g.149782207::chr6:g.117646467) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-003 6970-T01-IM6-69	P-0036 970-T01-IM6		ROS1-CD74	CD74	out-of-frame	three_prime	117650273	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q33.1;q22.1)(chr5:g.149783480::chr6:g.117650273) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-000 0632-T01-IM3-70	P-0000 632-T01-IM3		ROS1-CD74	CD74	in-frame	three_prime	117646313	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		NA Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 2722-T01-I M6-71	P-0032 722-T0 1-IM6		ROS1 -CD7 4	CD7 4	unknown	threonine_prime	117645 677	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782874:chr6:g.117645677) Protein Fusion: mid-exon {CD74:ROS1}	NA
EVT-P-001 4506-T01-I M6-72	P-0014 506-T0 1-IM6		ROS1 -CD7 4	CD7 4	unknown	threonine_prime	117650 623	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 32-43): t(5;6)(q32;q22.1)(chr5:g.149782871:chr6:g.117650623) Protein fusion: mid-exon (CD74:ROS1)	NA
EVT-P-001 8222-T01-I M6-73	P-0018 222-T0 1-IM6		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117646 209	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783915:chr6:g.117646209) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-000 1437-T01-I M3-74	P-0001 437-T0 1-IM3		ROS1 -CD7 4	CD7 4	out-of-frame	threonine_prime	117649 412	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		None Protein fusion: out of frame (CD74-ROS1)	NA
EVT-P-001 8513-T01-I M6-75	P-0018 513-T0 1-IM6		ROS1 -CD7 4	CD7 4	unknown	threonine_prime	117647 544	33	1800	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 33 to 43): t(5;6)(q32;q22.1)(chr5:g.149783804:chr6:g.117647544) Protein Fusion: mid-exon {CD74:ROS1}	NA
EVT-P-000 0569-T01-I M3-76	P-0000 569-T0 1-IM3		ROS1 -CD7 4	CD7 4	out-of-frame	five_prime	117647 622	33	1789	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		NA Protein fusion: out of frame (ROS1-CD74)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 5107-T01-I M6-77	P-0025 107-T0 1-IM6		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117647 170	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782204::chr6:g.117647170) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-000 0840-T01-I M3-78	P-0000 840-T0 1-IM3		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117646 938	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		ROS1 (NM_002944) - CD74 (NM_001025158) Translocation: t(6;5)(q22.1;q32)(chr6:g.117646938::chr5:g.149783508) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-000 7108-T01-I M5-79	P-0007 108-T0 1-IM5		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117646 132	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1 to 6 fused with ROS1 exons 34 to 43) : t(5;6)(chr5q32;chr6q22.1)(chr5:g.149783704::chr6:g.117646132) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-002 7196-T01-I M6-80	P-0027 196-T0 1-IM6		ROS1 -CD7 4	CD7 4	out-of-frame	threonine_prime	117650 377	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149782321::chr6:g.117650377) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-002 6589-T02-I M6-81	P-0026 589-T0 2-IM6		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117646 032	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) Fusion (CD74 exon6 fused to ROS1 exon34) : t(5;6)(q32;q22.1)(chr5:g.149782990::chr6:g.117646032) Protein Fusion: in frame (CD74:ROS1)	NA
EVT-P-002 1054-T01-I M6-82	P-0021 054-T0 1-IM6		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117647 162	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783021::chr6:g.117647162) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 8134-T01-I M5-83	P-0008 134-T0 1-IM5		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117646 080	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD274 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(5q32;q22.1)(chr5:g.149783121::chr6:g.117646080) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-002 3617-T01-I M6-84	P-0023 617-T0 1-IM6		ROS1 -CD7 4	CD7 4	in-frame	threonine_prime	117646 444	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CD74 (NM_00102515) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782946::chr6:g.117646444) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-006 5104-T01-I M7-86	P-0065 104-T0 1-IM7		ROS1 -CD7 4	CD7 4	in-frame	five_prime	117646 386	34	1853	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		ROS1 (NM_002944) - CD74 (NM_001025159) rearrangement: t(5;6)(q32;q22.1)(chr5:g.149783694::chr6:g.117646386) Protein Fusion: in frame {ROS1:CD74}	NA
EVT-P-004 1118-T07-I M7-87	P-0041 118-T0 7-IM7		ROS1 -CD7 4	CD7 4	out-of-frame	five_prime	117648 241	33	1789	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		ROS1 (NM_002944) - CD74 (NM_001025159) rearrangement: t(5;6)(q32;q22.1)(chr5:g.149782213::chr6:g.117648241) Protein Fusion: out of frame {ROS1:CD74}	NA
EVT-P-000 8996-T04-I M6-88	P-0008 996-T0 4-IM6		ROS1 -CEP 85L	CEP 85L	unknown	threonine_prime	117687 428	18	875	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		CEP85L (NM_001178035) - ROS1 (NM_002944) rearrangement : c.242-19954:CEP85L_c.2623:ROS1del Protein Fusion: mid-exon {CEP85L:ROS1}	NA
EVT-P-005 5135-T03-I M7-90	P-0055 135-T0 3-IM7		ROS1 -COL 13A1	COL 13A1	in-frame	five_prime	117658 665	31	1694	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		ROS1 (NM_002944) - COL13A1 (NM_001130103) rearrangement: t(6;10)(q22.1;q22.1)(chr6:g.117658665::chr10:g.71715849) Protein Fusion: in frame {ROS1:COL13A1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 0458-T01-I M6-98	P-0020 458-T0 1-IM6		ROS1 -EZR	EZR	in-frame	threonine_prime	117646 033	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 fused with ROS1 exons 34-43): c.1090+207:ER_c.5558-455:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-000 6047-T02-I M5-99	P-0006 047-T0 2-IM5		ROS1 -EZR	EZR	in-frame	threonine_prime	117647 241	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 with ROS1 exons 34-43): c.1091-355:EZR_c.5557+146:ROS1del Protein fusion: in frame (EZR-ROS1)	NA
EVT-P-000 1117-T01-I M3-100	P-0001 117-T0 1-IM3		ROS1 -EZR	EZR	unknown	threonine_prime	117650 595	32	1755	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) Deletion: c.1090+214_c.5263del Protein fusion: mid-exon (EZR-ROS1)	NA
EVT-P-001 0989-T01-I M5-101	P-0010 989-T0 1-IM5		ROS1 -EZR	EZR	out-of-frame	threonine_prime	117649 729	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 fused with ROS1 exons 33-43): c.1090+215:ER_c.5366+763:ROS1del Protein fusion: out of frame (EZR-ROS1)	NA
EVT-P-000 6047-T03-I M6-102	P-0006 047-T0 3-IM6		ROS1 -EZR	EZR	in-frame	threonine_prime	117647 241	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1091-356:EZR_c.5557+146:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-000 3970-T01-I M5-103	P-0003 970-T0 1-IM5		ROS1 -EZR	EZR	out-of-frame	threonine_prime	117649 698	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		ROS1 (NM_002944) - EZR (NM_003379) rearrangement: c.1090+142 C:EZR_c.5366+794T:ROS1del Protein fusion: out of frame (EZR-ROS1)	NA
EVT-P-001 4437-T01-I M6-104	P-0014 437-T0 1-IM6		ROS1 -EZR	EZR	out-of-frame	threonine_prime	117650 363	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		ROS1 (NM_002944) _EZR (NM_003379) rearrangement: c.1090+229:ER_c.5366+129:ROS1del Protein fusion: out of frame (EZR-ROS1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 4609-T01-I M6-105	P-0034 609-T0 1-IM6		ROS1 -EZR	EZR	in-frame	threonine_prime	117647 244	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1090+175:EZR_c.5557+143:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-001 4304-T02-I M6-106	P-0014 304-T0 2-IM6		ROS1 -EZR	EZR	in-frame	threonine_prime	117645 949	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: (EZR exons 1 to 9 fused with ROS1 exons 34 to 43): c.1090+79:EZR_c.5558-371:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-000 4445-T01-I M5-107	P-0004 445-T0 1-IM5		ROS1 -GOLGB1	GOLGB1	in-frame	threonine_prime	117645 253	34	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		GOLGB1 (NM_004487) - ROS1 (NM_002944) rearrangement : t(3;6)(3q13.33;6q22.1)(chr3:121434446::chr6:117645253) Protein fusion: in frame (GOLGB1-ROS1)	NA
EVT-P-000 5086-T01-I M5-108	P-0005 086-T0 1-IM5		ROS1 -GOPC	GOPC	in-frame	threonine_prime	117644 378	34	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		GOPC(NM_020399) -ROS1(NM_002944) - Fusion (GOPC exon 8 fused to ROS1 exon 34) : c.1259-1533:GOPC_c.5641+1117:ROS1del Protein fusion: in frame (GOPC-ROS1)	NA
EVT-P-000 7626-T01-I M5-109	P-0007 626-T0 1-IM5		ROS1 -GOPC	GOPC	in-frame	threonine_prime	117644 860	34	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		GOPC (NM_020399) - ROS1 (NM_002944) fusion (GOPC exons 1-8 fused with ROS1 exons 35-43) : c.1259-1217: GOPC_c.5641+635:ROS1del Protein fusion: in frame (GOPC-ROS1)	NA
EVT-P-001 9111-T01-I M6-111	P-0019 111-T0 1-IM6		ROS1 -GOPC	GOPC	out-of-frame	five_prime	117643 452	35	1881	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		ROS1 (NM_002944) - GOPC (NM_020399) rearrangement: c.5642-895: ROS1_c.912+138:GOPCdup Protein Fusion: out of frame {ROS1:GOPC}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 8302-T02-I M7-112	P-0058 302-T0 2-IM7		ROS1 -GRIK2	GRIK2	unknown	five_prime	117622188	42	2228	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Fibronectin type III domain			ROS1 (NM_002944) - GRIK2 (NM_021956) fusion: c.6682:ROS1_c.2086-17008:GRIK2inv Protein Fusion: mid-exon {ROS1:GRIK2}	NA
EVT-P-002 1914-T01-I M6-113	P-0021 914-T0 1-IM6		ROS1 -LARS1	LARS1	out-of-frame	five_prime	117646519	33	1853	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		ROS1 (NM_002944) - LARS (NM_020117) rearrangement: t(5;6)(q32,q22.1)(chr5:g.145500267::chr6:g.117646519) Protein Fusion: out of frame {ROS1:LARS}	NA
EVT-P-003 6702-T01-I M6-118	P-0036 702-T0 1-IM6		ROS1 -NUS1	NUS1	unknown	three_prime	117711069	12	421	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		ROS1 (NM_002944) rearrangement: c.1263-60:ROS1_chr6:g.118032766inv Transcript Fusion {NUS1:ROS1}	NA
EVT-P-001 6046-T01-I M6-171	P-0016 046-T0 1-IM6		ROS1 -SLC34A2	SLC34A2	out-of-frame	three_prime	117662059	30	1693	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		ROS1 (NM_002944) - SLC34A2(NM_006424) fusion: t(4;6) (p15;q22)(chr4:g.25667671::chr6:g.117662059) Protein Fusion: out of frame {SLC34A2:ROS1}	NA
EVT-P-001 0984-T01-I M5-175	P-0010 984-T0 1-IM5		ROS1 -SLC4A4	SLC4A4	in-frame	three_prime	117646719	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SLC4A4 (NM_001134742) - ROS1 (NM_002944) rearrangement: t(4;6)(q13.3;q22.1)(chr4:g.72429843::chr6:g.117646719) Protein fusion: in frame (SLC4A4-ROS1)	NA
EVT-P-001 3451-T01-I M5-178	P-0013 451-T0 1-IM5		ROS1 -STX7	STX7	out-of-frame	three_prime	117649456	32	1789	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		STX7 (NM_003569) - ROS1 (NM_002944) rearrangement: c.86-3411:STX7_c.5366+1036:ROS1del Protein fusion: out of frame (STX7-ROS1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 9670-T03-I M6-180	P-0019 670-T0 3-IM6		ROS1 -TFG	TFG	in-frame	threonine_prime	117643761	35	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		TFG (NM_001195478) - ROS1 (NM_002944) fusion (TFG exon 4 fused to ROS1 exon 35): t(3;6)(q12.2;q22.1) (chr3:g.100448554::chr6:g.17643761) Protein Fusion: in frame {TFG:ROS1}	NA
EVT-P-001 9670-T02-I M6-181	P-0019 670-T0 2-IM6		ROS1 -TFG	TFG	in-frame	threonine_prime	117643761	35	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		TFG (NM_001195478) - ROS1 (NM_002944) fusion (TFG exon 4 fused to ROS1 exon 35): t(3;6)(q12.2;q22.1) (chr3:g.100448554::chr6:g.17643761) Protein Fusion: in frame {TFG:ROS1}	NA
EVT-P-000 3207-T01-I M5-183	P-0003 207-T0 1-IM5		ROS1 -TPM3	TPM3	in-frame	threonine_prime	117642993	35	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		NA Protein fusion: in frame (TPM3-ROS1)	NA
EVT-P-002 8865-T02-I M6-184	P-0028 865-T0 2-IM6		ROS1 -TPM3	TPM3	in-frame	threonine_prime	117643589	35	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		TPM3 (NM_001278190) - ROS1 (NM_002944) fusion (TPM3 exons 1-6 with ROS1 exons 35-43): t(1;6)(q21.3;q22.1)(chr1:g.154133648::chr6:g.117643589) Protein Fusion: in frame {TPM3:ROS1}	NA
EVT-P-000 1840-T01-I M3-185	P-0001 840-T0 1-IM3		SDC4 -ROS1	SDC4	in-frame	threonine_prime	117650688	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) Translocation: t(20;6)(q13.12;q22.1)(chr20:43962892::chr6:117650688) Protein fusion: in frame (SDC4-ROS1)	NA
EVT-P-000 1840-T02-I M5-186	P-0001 840-T0 2-IM5		SDC4 -ROS1	SDC4	in-frame	threonine_prime	117650732	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(20;6)(q13.12;q22.1) (chr20:g.43962893::chr6:g.117650732) Protein fusion: in frame (SDC4-ROS1)	NA
EVT-P-005 4419-T02-I M6-187	P-0054 419-T0 2-IM6		SDC4 -ROS1	SDC4	unknown	threonine_prime	117645734	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645734::chr20:g.43955928) Protein Fusion: mid-exon {SDC4:ROS1}	NA

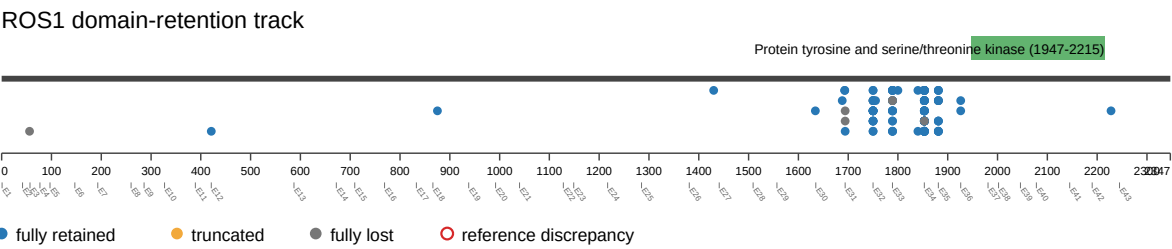
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 4419-T01-I M6-188	P-0054 419-T0 1-IM6		SDC4-ROS1	SDC4	unknown	threep_prime	117645734	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645734::chr20:g.43955928) Protein Fusion: mid-exon {SDC4:ROS1}	NA
EVT-P-004 7840-T02-I M6-189	P-0047 840-T0 2-IM6		SDC4-ROS1	SDC4	unknown	threep_prime	117645871	34	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645871::chr20:g.43955947) Protein Fusion: mid-exon {SDC4:ROS1}	NA
EVT-P-000 4414-T01-I M5-190	P-0004 414-T0 1-IM5		SDC4-ROS1	SDC4	in-frame	threep_prime	117651952	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-4 fused with ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117651952::chr20:g.43957039) Protein fusion: in frame (SDC4-ROS1)	NA
EVT-P-004 3938-T01-I M6-191	P-0043 938-T0 1-IM6		SDC4-ROS1	SDC4	in-frame	threep_prime	117651208	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117651208::chr20:g.43962372) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-000 4946-T02-I M5-192	P-0004 946-T0 2-IM5		SDC4-ROS1	SDC4	in-frame	threep_prime	117651013	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117651013::chr20:g.43962011) Protein fusion: in frame (SDC4-ROS1)	NA
EVT-P-003 8394-T01-I M6-193	P-0038 394-T0 1-IM6		SDC4-ROS1	SDC4	in-frame	threep_prime	117654360	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) rearrangement: t(6;20)(q22.1;q13.12)(chr6:g.117654360::chr20:g.43961742) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-002 7630-T01-I M6-194	P-0027 630-T0 1-IM6		SDC4-ROS1	SDC4	in-frame	threep_prime	117651891	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117651891::chr20:g.43962429) Protein Fusion: in frame {SDC4:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 4541-T01-I M7-195	P-0064 541-T0 1-IM7		SDC4-ROS1	SDC4	in-frame	threonine_prime	117650669	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117650669::chr20:g.43962537) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-002 2578-T02-I M6-196	P-0022 578-T0 2-IM6		SDC4-ROS1	SDC4	in-frame	threonine_prime	117652158	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117652158::chr20:g.43962266) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-002 2578-T01-I M6-197	P-0022 578-T0 1-IM6		SDC4-ROS1	SDC4	in-frame	threonine_prime	117652158	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117652158::chr20:g.43962266) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-001 2008-T01-I M5-198	P-0012 008-T0 1-IM5		SDC4-ROS1	SDC4	in-frame	threonine_prime	117652119	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) reciprocal fusion (SDC4 exons 1-2 with ROS1 exons 32-43): t(6;20)(q22.1;q13.12) (chr6:g.117652119::chr20:g.43964016) Protein fusion: in frame (SDC4-ROS1)	NA
EVT-P-001 7325-T01-I M6-199	P-0017 325-T0 1-IM6		SDC4-ROS1	SDC4	in-frame	threonine_prime	117654375	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117654375::chr20:g.43963860) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-001 2008-T02-I M6-200	P-0012 008-T0 2-IM6		SDC4-ROS1	SDC4	in-frame	threonine_prime	117652099	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) reciprocal fusion (SDC4 exons 1-2 with ROS1 exons 32-43): t(6;20)(q22.1;q13.12) (chr6:g.117652119::chr20:g.43964016) Protein Fusion: in frame {SDC4:ROS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 6504-T01-I M6-201	P-0016 504-T0 1-IM6		SDC4-ROS1	SDC4	in-frame	threep_rime	117650681	32	1750	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117650681::chr20:g.43961904) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-000 3507-T01-I M5-202	P-0003 507-T0 1-IM5		SLC16A10-ROS1	SLC16A10	in-frame	five_p_rime	117739561	2	56	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Fibronectin type III domain		NA Protein fusion: in frame (ROS1-SLC16A10)	NA
EVT-P-004 2481-T01-I M6-203	P-0042 481-T0 1-IM6		SLC4A4-ROS1	SLC4A4	out-of-frame	threep_rime	117646662	33	1853	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		SLC4A4 (NM_001134742) - ROS1 (NM_002944) fusion: t(4;6)(q13.3;q22.1)(chr4:g.72430718::chr6:g.117646662) Protein Fusion: out of frame {SLC4A4:ROS1}	NA
EVT-P-004 5322-T02-I M6-205	P-0045 322-T0 2-IM6		TFG-ROS1	TFG	in-frame	threep_rime	117644683	34	1881	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Fibronectin type III domain		TFG (NM_001195478) - ROS1 (NM_002944) fusion: t(3;6)(q12.2;q22.1)(chr3:g.100448076::chr6:g.117644683) Protein Fusion: in frame {TFG:ROS1}	NA
EVT-P-000 7870-T02-I M6-207	P-0007 870-T0 2-IM6		ZNF157-ROS1	ZNF157	unknown	five_p_rime	117659405	31	1694	True	lost	0.0	False	Fibronectin type III domain	Protein tyrosine and serine/threonine kinase		ROS1 (NM_002944) Rearrangement : t(X;6)(p11.23;q22.1)(chrX:g.47262831; chr6:g.117659405) Antisense Fusion	NA

Figures

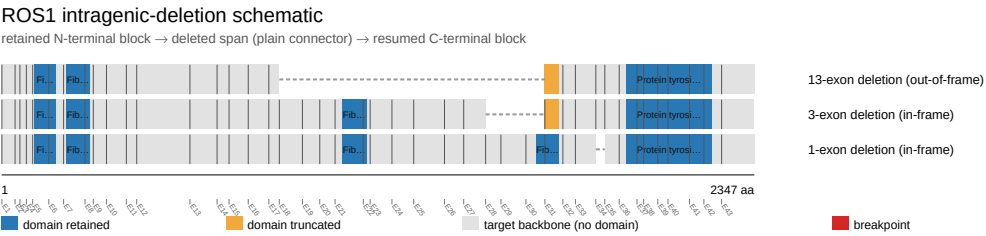
domain retention outliers



fusion schematic



intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

